1. Consider a system of the form $\dot{X} = AX$. Determine which of the systems are stable/marginally stable/unstable.

(a) $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

(b) $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

(c) $A = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}$

Gavin Kelley

5411 - Assignment 3

(a) $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, diagonalize: $\det(A - \lambda I) = 0$

$$\det \begin{bmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 0 & 0 & -\lambda \end{bmatrix} = 0$$

$$-\lambda(\lambda^2 - 0) + (0) + \cancel{0(\ldots)}^{0} = 0$$

$$-\lambda^3 = 0 \Rightarrow \boxed{\text{System Unstable}}$$

(repeated eigenvalues on $j\omega$-axis)

(b) $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$$\det \begin{pmatrix} -\lambda & 1 \\ -1 & -\lambda \end{pmatrix} = 0$$

$$\lambda^2 + 1 = 0 \Rightarrow \lambda = \pm i \Rightarrow \boxed{\text{System is marginally stable}}$$

$(\mathrm{Re}[\lambda] = 0)$

(c) $A = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}$

$$\det \begin{pmatrix} -\lambda & 1 \\ -1 & 1-\lambda \end{pmatrix} = 0$$

$$-\lambda(1-\lambda) + 1 = 0$$

$$\lambda^2 - \lambda + 1 = 0 \Rightarrow \lambda = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1}{2} \pm i\frac{\sqrt{3}}{2}$$

$\mathrm{Re}[\lambda] > 0 \Rightarrow \boxed{\text{System unstable}}$

2. Consider the following system

$$\dot{X} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} X + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t)$$

We want to design a state feedback controller of the form $u(t) = -KX(t) + v(t)$ to stabilize the system. Find K such that the system has (repeated) eigenvalues at -1 and -1.

$$\dot{X} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} X + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \left(-KX + v(t) \right)$$

$n=2$ $\quad$ $m=1$

$K \in \mathbb{R}^{m \times n}$

let $k = \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}^T$, $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

then $u = -\begin{bmatrix} k_1 \\ k_2 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -\begin{bmatrix} k_1 & k_2 \end{bmatrix} X$

$$\Rightarrow \dot{X} = \left(\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} k_1 & k_2 \end{bmatrix} \right) X$$

$$\dot{X} = \begin{bmatrix} -k_1 & 1-k_2 \\ -1 & 0 \end{bmatrix} X$$

$\lambda = -1$, order 2

$$\det\left(\begin{bmatrix} -k_1 & 1-k_2 \\ -1 & 0 \end{bmatrix} - \lambda I \right) = 0, \quad \lambda = -1$$

$$\det\begin{pmatrix} 1-k_1 & 1-k_2 \\ -1 & 1 \end{pmatrix} = 0$$

$$(1-k_1) - (-1)(1-k_2) = 0$$

$$k_1 + k_2 = 2$$

Generally, $\det\begin{bmatrix} -\lambda-k_1 & 1-k_2 \\ -1 & -\lambda \end{bmatrix} = 0$

$$-\lambda(-\lambda-k_1) + (1-k_2) = 0$$

$$\lambda^2 + \lambda k_1 + (1-k_2) = 0$$

$$\lambda = \frac{-k_1}{2} \pm \frac{\sqrt{k_1^2 - 4(1-k_2)}}{2} = -1$$

MATLAB Syms solve()
(attached)

$$\Rightarrow \quad k_1 = 2, \quad k_2 = 0 \qquad \left(\text{imag}[\lambda] = \frac{\sqrt{(4)-(4-0)}}{2} = 0 \right)$$

$$\therefore \quad \boxed{K = [k_1 \;\; k_2] = [2, 0]}$$

```
syms k1 k2;

l1 = (-k1/2) + 0.5*sqrt(k1^2-4*(1-k2)) == -1;
l2 = (-k1/2) - 0.5*sqrt(k1^2-4*(1-k2)) == -1;

S = solve(l1,l2,[k1 k2]);

Sk1 = double(S.k1)
```

```
Sk1 = 2
```

```
Sk2 = double(S.k2)
```

```
Sk2 = 0
```

```
% sanity check
k1 = 2; k2 = 0;

A = [0 1; -1 0] - [1;0]*[k1 k2]
```

```
A = 2×2
    -2     1
    -1     0
```

```
[v,D] = eig(A)
```

```
v = 2×2
    0.7071   -0.7071
    0.7071   -0.7071
D = 2×2
   -1.0000         0
         0   -1.0000
```

ECGR 4111/5411: Control Systems Theory I

Assignment 3
Marks: 100

Unless mentioned otherwise, each student must solve **all** the problems. You must do all the subproblems for a problem to get full credit (Bonus credit (sub)problems are optional). Show all the steps and associated calculations as well for full credit. For any of the problems, you are allowed to use any numerical tools. However, you must submit your code with your submission.

1. Consider a system of the form $\dot{X} = AX$. Determine which of the systems are stable/marginally stable/unstable.
   (a) $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$
   (b) $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
   (c) $A = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}$
2. Consider the following system

$$\dot{X} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} X + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t)$$

   We want to design a state feedback controller of the form $u(t) = -KX(t) + v(t)$ to stabilize the system. Find K such that the system has (repeated) eigenvalues at -1 and -1.

   The problems below are not graded.

3. (a) Revisit the following problem from Assignment 1:

$$\frac{\mathrm{d}^4 y}{\mathrm{d}t^4} + 7\frac{\mathrm{d}^3 y}{\mathrm{d}t^3} + 6\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} - 9\frac{\mathrm{d}y}{\mathrm{d}t} = 2U(t).$$

   Check whether the system is stable/marginally stable/unstable.

   (b) Revisit the following problem from Assignment 1:

$$\frac{\mathrm{d}^n y}{\mathrm{d}t^n} = -y.$$

   Check whether the system is stable/marginally stable/unstable.

4. Consider the system

$$\dot{X} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{A} X + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} u(t)$$

   (a) Design a state-feedback controller $u(t) = -KX(t)$ to place the eigenvalues at -1 and -2.

(b) Once you have found K from part (a), you have the system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = (A - BK) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

plot $x_1(t)$ and $x_2(t)$ vs. t.

(c) Repeat part (a) to place the eigenvalues at -3 and -4.

(d) Repeat part (b) with the K found from part (c).

(e) Which pair of eigenvalues (part (a) or part (c)) makes the system (x_1 and x_2) go to zero faster? What can you infer about the connection between the eigenvalues of the $A - BK$ matrix and the speed of convergence of x_1, x_2 to zero.